Bowen Jing

Arthur12137 linkedin github Beijing, China +86 18321700541 Personal Page Google Scholar

RESEARCH VISION

Bowen Jing works on **Physical AI** — intelligence grounded in physical interaction. His focus spans three coupled problems: building physical simulation environments faithful enough to be trusted, giving agents the observations that physical contact actually provides — tactile above all — and studying which embodied algorithms, VLA policies and world-action models, become possible once both exist. He believes the bottleneck is no longer policy architecture, but the environments and the senses we build for them.

EDUCATION

University of Manchester: ***MSc.*** *Advanced Computer Science* - ***Grade:*** *Distinction (76)* — Sept. 2023 - Sept. 2024

University of Manchester: ***BSc.*** *Computer Science* — 2020 - 2023

Honours & Awards: University of Manchester Kilburn Entry Scholarship (awarded for A*A*A* at entry) • Academic Representative, ACS Artificial Intelligence (2023–2024)

PUBLICATIONS

1. **Bowen Jing**. StyleDrive: Driving-Style Aware Benchmarking for End-to-End Autonomous Driving. **AAAI 2026 (Oral)**. [link]
2. **Bowen Jing**. SoftVTBench: A Visuo-Tactile Benchmark for Safe Deformable-Object Manipulation. In review, **AAAI 2027**. [link]
3. **Bowen Jing**. CounterScene: Counterfactual Causal World Models for Safety-Critical Scenario Generation. **ECCVW 2026**. [link]

RESEARCH EXPERIENCE

SoftVTBench: A Visuo-Tactile Benchmark for Safe Deformable-Object Manipulation (AAAI 2027 Submission) [link]:

- **Led** the construction of a **visuo-tactile benchmark** for **safe manipulation of soft and deformable objects**, organized around the **Goal–Safety Gap** — the trade-off between task success and deformation safety — with a 2×2 task suite (spatial/object $\times$ rigid/soft).
- **Built** 10 classes of calibrated soft-body assets in **Isaac Sim** FEM simulation, and proposed a **contact-patch strain deformation-safety metric** together with a **Safe Success** evaluation protocol.
- **Developed** a large-scale simulation data pipeline, collecting 3,000+ rigid/soft demonstration trajectories across a multi-machine cluster, and trained and evaluated **Pi0.5, ACT, and Diffusion Policy** under a unified protocol.
- **Implemented** a vectorized parallel evaluation framework and an **OOD generalization protocol**, enabling systematic comparison of vision-only against vision-plus-tactile policies; driving the dataset/code release and the IROS 2026 Challenge.

CounterScene: Counterfactual Causal World Models for Safety-Critical Scenario Generation (ECCVW 2026) [link]:

- **Developed CounterScene**, a novel framework for **safety-critical scenario generation** via structured counterfactual reasoning in generative BEV world models.
- **Reformulated** scenario generation as a **minimal counterfactual intervention problem**, enabling controllable and interpretable modification of multi-agent behaviors under fixed scene conditions.
- **Designed** a **Causal Interaction Graph (CIG)** to explicitly model dynamic inter-agent dependencies, supporting structured propagation of behavioral changes during closed-loop simulation.
- **Proposed** a conflict-aware **causal agent identification mechanism** that selects critical agents via geometric interaction analysis, outperforming heuristic and TTC-based strategies.
- **Introduced** stage-adaptive counterfactual diffusion guidance, decomposing safety constraints into spatial and temporal components to generate minimal yet effective interventions.
- **Achieved** state-of-the-art performance on **nuScenes**, increasing long-horizon collision rate (12.3% $\rightarrow$ 22.7%) while maintaining superior trajectory realism (ADE 1.88 vs. 2.09).
- **Demonstrated** strong zero-shot generalization to **nuPlan**, validating robustness across diverse driving domains.
- **Positioned** CounterScene as a bridge between **generative world models and causal reasoning**, enabling interpretable and interaction-aware closed-loop evaluation.

StyleDrive: Driving-Style Aware Benchmarking for End-to-End Autonomous Driving (AAAI 2026, Oral) [link]:

- **Developed StyleDrive**, the **first real-world large-scale dataset** for personalized end-to-end autonomous vehicles.
- **Constructed** over 30,000 real-world driving scenarios enriched with fine-grained annotations for A, N, C driving styles, derived through a hybrid pipeline integrating HD map topology, motion heuristics, and fine-tuned Video-LLaMA3.
- **Designed** a multi-stage annotation framework combining objective behavior analysis and subjective VLM-based reasoning, with human-in-the-loop validation for high-quality, interpretable style labels.
- **Introduced** a novel evaluation metric — **Style-Modulated PDMS (SM-PDMS)**.
- **Bridged** the Vision-Language-Action paradigm to real-world autonomous driving by conditioning action generation on linguistic style descriptors, demonstrating how language-guided modulation enhances perception-to-control reasoning.
- **Positioned** StyleDrive as an embodied VLA benchmark, where vision encoders, language-based style priors, and trajectory decoders jointly enable personalized and interpretable driving behaviors.

Sentence-Level Relation Extraction [link]: Technologies: PA-LSTM, Bi-LSTM, C-GCN, BERT, RoBERTa, SpanBERT

- **Implemented and analyzed** a suite of neural architectures for sentence-level relation extraction, comparing sequential (PA-LSTM, Bi-LSTM), structural (C-GCN), and pretrained transformer-based models (BERT, RoBERTa, SpanBERT).
- **Designed a novel entity marker injection strategy** for BERT-based models, improving over standard entity masking by enabling explicit modeling of entity-pair interactions via learned token embeddings.
- **Engineered multiple experimental conditions** on three benchmark datasets (TACRED, Re-TACRED, TACREV), including:
  - Constructing controlled subsets of sentences with varying lengths (short: $\leq$15 tokens, medium: 16–30, long: $>$30), revealing transformer robustness in syntactically complex cases.
  - Building reduced training sets (10%, 25%, 50%) to benchmark few-shot capabilities.
- **Conducted rigorous ablation studies** across entity representation types (e.g., hidden state pooling vs. special token concatenation), and input granularity, to disentangle the sources of performance gains in pretrained models.
- **Led a 3-person team**, defining milestones and managing deliverables, while contributing key technical insights.
- **Achieved top-5 performance in class**, scoring 92.75% with a well-justified analysis report, praised for its methodological rigor and clarity of comparison across paradigms.

Comparative Analysis of Deep Learning and Traditional Computer Vision in Robotic Vision [link]: Technologies: ResNet, Vision Transformers (ViT), Bag-of-Visual-Words (BoVW), SIFT, K-Means, SVM April 2024–May 2024

- **Conducted a comprehensive comparative study** between traditional handcrafted feature pipelines and modern deep learning-based vision architectures, targeting image classification for robotic perception tasks.
- **Implemented and optimized** multiple models including ResNet and Vision Transformers (ViT), trained from scratch and fine-tuned with ImageNet21k weights, to evaluate learned feature representations.
- **Engineered a classical BoVW pipeline** from the ground up using SIFT descriptors, K-Means clustering for visual vocabulary generation, and SVM classification, offering a strong baseline for feature-based vision.
- **Designed and curated noise-augmented datasets** derived from CIFAR-100 and Caltech-101, injecting synthetic Gaussian noise at three levels (low/medium/high) to rigorously assess model robustness under degraded conditions.
- **Analyzed model behavior** through confusion matrices, class-wise accuracy breakdowns, and training dynamics, developing insight into how attention-based architectures handle inter-class similarity compared to local-descriptor-based methods.

PROFESSIONAL EXPERIENCE

Tuojing Intelligence Aug. 2025 – Present
Tech Lead Beijing, China

- **Visuo-Tactile Ecosystem:** Built the group's visuo-tactile research stack end to end — simulation assets, tactile sensing pipelines, and evaluation protocols for safe manipulation of deformable objects.
- **Embodied Algorithms:** Trained, reproduced, and benchmarked large-scale embodied policies including Pi0/Pi0.5/Pi0.6, Fast-WAM, LingBot-VLA, and DreamZero.
- **Generative Simulation:** Developed diffusion-based generative simulators for closed-loop embodied decision-making and interactive environment modeling.
- **AI Infrastructure:** Maintained HPC clusters and multi-GPU training pipelines, ensuring scalable, stable, and reproducible large-scale experiments.

Tsinghua University Feb. 2025 – Aug. 2025
Research Intern, Institute for AI Industry Research (AIR) Beijing, China

- **Behavior & Style Modeling:** Disentangled driving style and physical signatures to enable personalized action generation.
- **VLM-grounded Control:** Integrated pre-trained VLMs into embodied control loops via hierarchical task decomposition.
- **Generative E2E Driving:** Explored generative V2A policies and multimodal scene understanding for robust decision-making.

KEY AI & DEEP LEARNING SKILLS

Programming & Frameworks: Python | PyTorch | TensorFlow | JAX | NumPy | CUDA | OpenCV | ONNX
Generative & Foundation Models: Diffusion Models | Transformers (ViT, LLaMA, BERT) | VAE | RAG | Prompt Engineering
Embodied & Autonomous Systems: CARLA | nuPlan | nuScenes | ROS | Isaac Sim / Isaac Lab | Gymnasium
Learning & Optimization: Contrastive Learning | Self-Supervised Learning | Causal Inference | Reinforcement Learning (D4PG, PPO)
Simulation & Evaluation: FEM Soft-Body Simulation | Closed-loop Evaluation | Behavior Cloning | Scenario Generation | Counterfactual Simulation
Tools & Research Practices: Linux | Weights&Biases | HuggingFace | Git | HPC Training | Dataset Design | Academic Writing